

Supplier Data Harmonization & Inventory Modeling Report

May 6, 2026

1. What I Built

A modular Python pipeline with the following files:

- **config.py**: API credentials
- **extractor.py**: Fetches data from European Sourcing API
- **transformer.py**: Cleans and standardises data
- **models.py**: Unified product schema
- **explorer.py**: Data exploration and quality analysis
- **query.py**: Query by category, stock, supplier
- **main.py**: Runs the full pipeline

2. Outcome

API Data Extraction

- Successfully connected to the European Sourcing Sandbox API.
- Implemented pagination logic using the "from" parameter to bypass the default limit of 52 products.
- Fetched 208 products for the dataset.

Data Cleaning & Standardisation

- Converted all product dimensions from mm to cm (value / 10).
- Standardised all text fields to lowercase and removed extra spaces.

Data Harmonisation

- **Category Mapping**: Created a unified taxonomy (**CATEGORY_MAP**) to group supplier categories into standard groups (e.g., "Razor" → "beauty").
- **Color Normalisation**: Built a **COLOR_MAP** to handle naming differences (e.g., "navy blue", "dark blue" → "blue").

3. Challenges & Solutions

Challenge	Impact	Solution
Incorrect API Header	API returned HTML instead of JSON.	Identified that <code>x-auth-token</code> was required instead of the standard <code>X-API-Key</code> via Swagger documentation.
Endpoint Misconfiguration	Used <code>/products</code> which was a web view.	Switched to the correct <code>POST /api/v1.2/search</code> endpoint for structured data.
Limited Fetch Size	API default was capped at 52 items.	Built a loop-based pagination system to skip records and collect a larger sample size.

5. Sandbox Limitations

The European Sourcing sandbox API returned data from **XD CONNECTS** supplier only. Other suppliers (Midocean, PF Concept, Clipper Interall, DATA NatuRe) were not available.

As a result, cross-supplier inconsistencies (e.g., "Colour" vs "Color") could not be demonstrated with real data. However, the pipeline is built to handle these cases via `COLOR_MAP` and `CATEGORY_MAP`. This limitation was communicated to the hiring team via email.

6. Assumptions

- All prices are in **EUR**
- All dimensions are in **mm** and were converted to **cm**.
- Missing fields are filled with **"unknown"** or **0** as defaults.
- Supplier identity is taken from the `supplier.name` field.

9. Technical Stack

- **Language:** Python 3.x
- **Output:** `unified_catalogue.json` + `unified_catalogue.csv`